# Portable Spec Kit — Spec-Persistent Development

## Spec-Persistent Development for AI-Assisted Engineering

A single portable file that defines your complete development methodology — personalized from the first session. Drop it into any project and the AI agent detects your GitHub profile, adapts to your expertise and working style, follows your engineering standards, manages tasks, writes tests, and maintains context across sessions.

SPECS → PLANS → TASKS → BUILD + TEST → RELEASES

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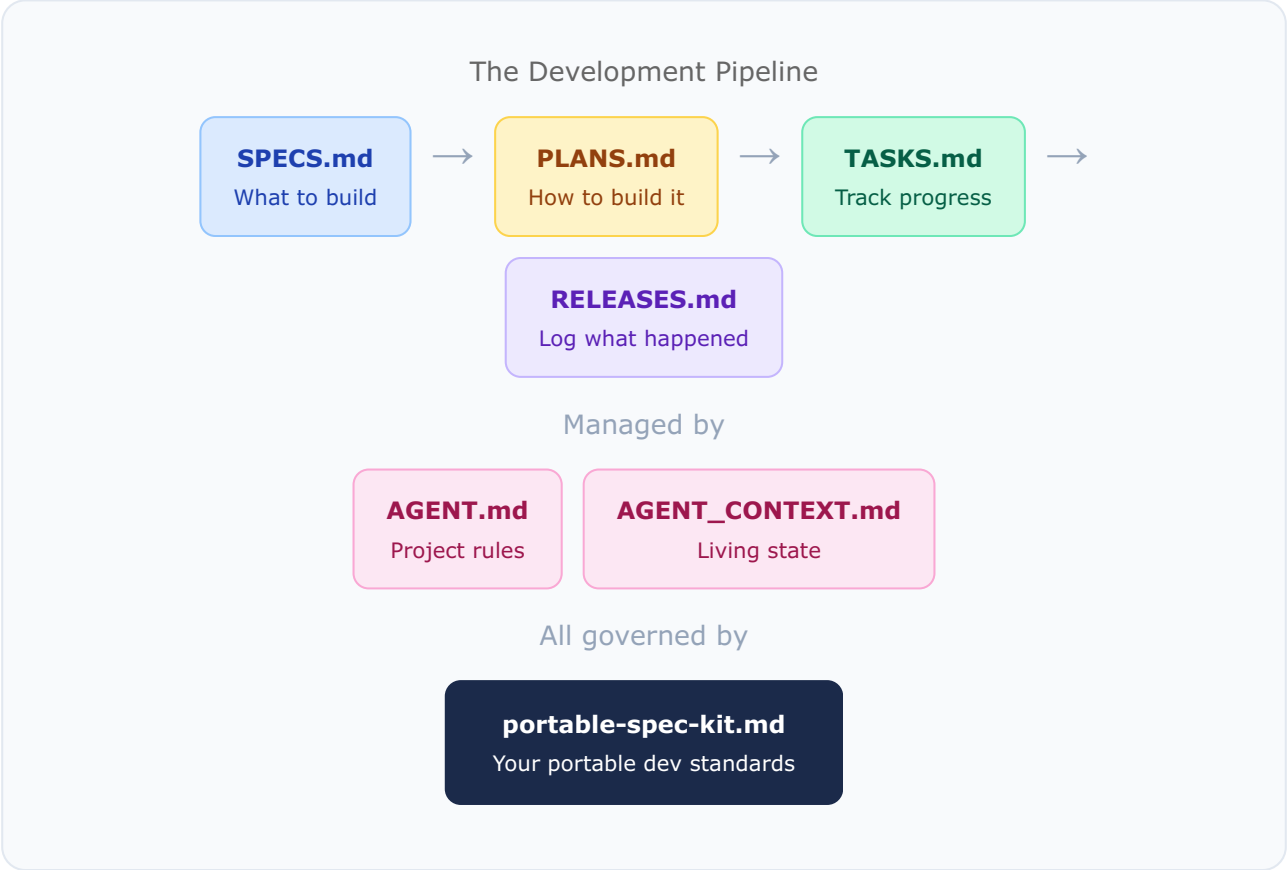
## The Framework

**Spec-Persistent Development** — specs always exist and stay current, but never block. The framework doesn't enforce a methodology — it supports waterfall, agile, or mixed. The AI agent maintains living specifications throughout development, adapting to how you work.




### 10 Core Strengths

| Strength | What It Means |
|---|---|
| 🪶 **Lightweight** | Single markdown file. Zero dependencies. Zero install. |
| 📦 **Portable** | One file → any repo. Works instantly. Symlinks for all agents. |
| 👤 **Personalized** | GitHub profile auto-detect. Adapts to your expertise, communication style, and working pattern. Tailored AI behavior. |
| 📋 **Spec-Persistent** | SPECS → PLANS → TASKS → RELEASES. Specs persist alongside your code. Your workflow, your choice. |

| | |
|---|---|
| 🤖 **Agent-Agnostic** | Claude, Copilot, Cursor, Windsurf, Cline — one source, all sync. |
| 🧠 **Context-Persistent** | AGENT_CONTEXT.md carries state across sessions. Pick up after weeks. Never lose context. |
| 🔒 **Secure** | Never expose API keys — even if asked. Strict .env rules. Code security enforced. |
| 🔄 **Non-Blocking** | Code first, specs later. Agent tracks silently. Never blocks the developer. |
| 🏗️ **Self-Scaffolding** | Auto-creates 6 agent files, README, workspace context. 8 stack templates built-in. Ready to code in seconds. |
| ✅ **Self-Validating** | Agent tests everything. Fixes before presenting. 98%+ coverage target. |

## What the Framework File Contains

The `portable-spec-kit.md` file is the single source of truth for your development methodology. It contains:

| Section | What It Governs |
|---|---|
| User Profile | Personalized AI — GitHub auto-detect, communication style, working pattern, AI delegation |
| Git & GitHub Rules | When to commit, push, ask before critical ops |
| Security Rules | Secret handling, .env practices, code security |
| Versioning | Two-level: framework v0.N.x (each publish) + release v0.N (milestones), auto-restructure on pull |
| Task Tracking | Add tasks FIRST, then work. Module-based organization. |
| Testing Standards | Coverage targets, edge case checklist, mock rules, self-validation |
| Code Quality | Review checklist, naming conventions, deployment checklist |
| Error Handling | Structured errors, logging, error boundaries, user-friendly messages |

| | |
|---|---|
| Branch & PR Workflow | Feature branches, PR format, squash merge, CI-before-merge requirement |
| CI & Community Contributions | GitHub Actions ci.yml template, stack-aware test commands, branch protection, PR workflow, contribution validation |
| Spec-Based Test Generation | Forward-flow stub generation, per-feature acceptance criteria format, stack-aware stubs, stub completion gate |
| Dependencies | Bundle size checks, lock files, audit, avoid unnecessary deps |
| Project Templates | 6 agent files + README + 8 source code structures (Web, Python, Mobile, Android, iOS, Full Stack, Full Stack + Mobile, Document) |
| Auto-Scan | Detects existing projects, creates missing files, restructures to standard — 13 work flow docs |
| Agent Behavior | Guide don't enforce, silent tracking, retroactive spec-filling, R→F→T traceability |

## Multi-Agent Support

One source file — `portable-spec-kit.md` — works with every AI coding agent via symlinks:

| Agent | File Created | How |
|---|---|---|
| **Claude Code** | `CLAUDE.md` | Symlink → portable-spec-kit.md |
| **GitHub Copilot** | `.github/copilot-instructions.md` | Symlink → portable-spec-kit.md |
| **Cursor** | `.cursorrules` | Symlink → portable-spec-kit.md |
| **Windsurf** | `.windsurfrules` | Symlink → portable-spec-kit.md |
| **Cline** | `.clinerules` | Symlink → portable-spec-kit.md |

**macOS/Linux:** Uses symlinks — edit `portable-spec-kit.md`, all agents read the update instantly.

**Windows:** Uses copies — re-run install command after editing, or let the AI agent keep files in sync.

## Complete Flow

1. **Install** — one command downloads `portable-spec-kit.md` + creates symlinks for all agents (Claude, Copilot, Cursor, Windsurf, Cline)

↓

2. **Open any AI agent — auto-scaffolds** — detects environment, creates WORKSPACE_CONTEXT.md, creates `agent/` with 6 files (AGENT, AGENT_CONTEXT, SPECS, PLANS, TASKS, RELEASES), project dirs (src/, tests/, docs/), README, .gitignore, .env.example, tests/test-release-check.sh

↓

3. **Work however you want** — "build me X" → agent adds to TASKS.md, builds, tests. "What should I do next?" → agent walks through spec-persistent flow. Never blocks.

↓

4. **Session ends** — agent updates AGENT_CONTEXT.md with progress, decisions, what's next

↓

5. **Come back later** — days, weeks, months. Agent reads AGENT_CONTEXT.md: "Here's where we left off." Continues exactly where you stopped.

↓

6. **Edit the framework** — update portable-spec-kit.md, all 5 agent symlinks read the change instantly. Standards evolve with you.

## Project Structure

```
your-project/
├── portable-spec-kit.md     ← Framework file (source)
├── .portable-spec-kit/      ← Kit config (user profiles, per-user)
│   └── user-profile/
│       └── user-profile-{username}.md
├── CLAUDE.md                ← Symlink (Claude Code)
├── .cursorrules             ← Symlink (Cursor)
```

```
├── WORKSPACE_CONTEXT.md    ← Auto-created (workspace state)
├── README.md               ← Auto-created (standard structure)
│
├── agent/                 ← Auto-created (project management)
│   ├── AGENT.md           ← Project rules, stack, brand
│   ├── AGENT_CONTEXT.md   ← Living state (updated every session)
│   ├── SPECS.md           ← Requirements & features
│   ├── PLANS.md        ← Architecture, methodology & research
│   ├── TASKS.md           ← Task tracking (module-based)
│   └── RELEASES.md         ← Version history
│
├── src/                   ← Your code
├── tests/                 ← Your tests
├── ard/                   ← Architecture docs (HTML + PDF)
└── ...
```

## The 6 Agent Files

| File | Purpose | When |
|---|---|---|
| `AGENT.md` | Project-specific instructions — stack, rules, brand, AI config | Setup |
| `AGENT_CONTEXT.md` | Living state — done, next, decisions, blockers, file structure | Every session |
| `SPECS.md` | What to build — requirements, features, acceptance criteria, scope | Before dev |
| `PLANS.md` | How to build — architecture, data model, phases, methodology, research, decision log | Before dev |
| `TASKS.md` | Module-based task tracking with progress summary table | During dev |
| `RELEASES.md` | Version changelog (categorized), test results, deployment log | End of version |

# How to Work in Spec-Persistent Development

## Phase 1: Specification

> **Goal:** Define WHAT you're building before touching code.

Start every project by writing `agent/SPECS.md`. This is a conversation between you and the AI:

| You Say | AI Does |
|---|---|
| "I want to build a resume editor" | Writes requirements, features table, scope boundaries, acceptance criteria in SPECS.md |
| "Add AI-powered bullet rewriting" | Adds feature to specs, estimates priority, updates scope |
| "That's out of scope for v1" | Moves to "Out of scope (future)" section |

**Rule:** No code is written until SPECS.md has requirements and acceptance criteria.

## Phase 2: Planning

> **Goal:** Define HOW you're building it — architecture, data model, phases.

AI writes `agent/PLANS.md` based on approved specs:

| Section | What AI Fills In |
|---|---|
| Stack | Recommends tech (frontend, backend, DB, hosting) with trade-offs. You approve. |
| Architecture | System design, component relationships, data flow |
| Directory Structure | Exact file/folder layout for the chosen stack |
| Data Model | Tables, types, relationships |
| API Endpoints | Method, path, description for each route |
| Build Phases | Phased delivery plan — foundation first, features later |

| | |
|---|---|
| Security | Auth, CORS, input validation, secret handling |

**Rule:** Deployment is deferred to release time. Don't discuss hosting during planning.

### Phase 3: Execution

> **Goal:** Build, test, and track every task.

AI creates `agent/TASKS.md` with module-based breakdown:

```
## Module 1: Project Setup
| # | Task | Status |
|---|------|:------:|
| 1.1 | Initialize Next.js + Tailwind | [x] |
| 1.2 | Create component structure | [ ] |

## Module 2: Core Features
| # | Task | Status |
|---|------|:------:|
| 2.1 | Build editor canvas | [ ] |
```

**Execution rules enforced by the framework:**

| Rule | What It Means |
|---|---|
| Tasks first | When you ask for something, AI adds it to TASKS.md BEFORE starting work |
| Self-validate | AI runs tests, fixes failures, only presents stable results |
| Test everything | Every feature gets tests — unit, integration, UI interaction |
| Edge cases | Empty inputs, boundaries, XSS, unicode, error paths — all tested |
| Coverage targets | Backend 98%+, frontend logic 98%+, UI components 85%+ |
| No broken features | User should NEVER discover something doesn't work |
| Context updated | AGENT_CONTEXT.md updated after significant work, after commits, and before any push |
| Flows updated | docs/work-flows/ updated if any work flow changed — flows always match code |

### Phase 4: Tracking

> **Goal:** Log what was delivered, tested, and deployed.

At end of each version, AI updates `agent/RELEASES.md`:

```
## v0.2 — AI Features (March 30, 2026)

### Summary
Added JD-tailored resume, PDF generation, 673 tests.

### Changes
- **Frontend:** JD Tailor modal, version manager, skills tags
- **Backend:** 3 PDF engines, expand API, fetch-url API
- **AI:** Bullet format detection, deduplication, length matching

### Tests
- 673 tests passing, 95% coverage

### Deployment
- Deployed to: aqibmumtaz.github.io
- Date: 2026-03-30
```

### Daily Workflow Summary

1. **Start session** — AI reads agent/AGENT.md + AGENT_CONTEXT.md. Knows exactly where you left off.

↓

2. **You give instructions** — "add dark mode", "fix scoring bug", "build PDF export"

↓

3. **AI adds to TASKS.md** first, then builds, tests, and shows results with coverage

↓

4. **You review** — approve, adjust, or ask for changes. AI fixes and re-tests.

↓

5. **You say "commit"** — AI commits with descriptive message. "push" sends to remote.

↓

6. **Session ends** — AI updates AGENT_CONTEXT.md. Next session picks up seamlessly.

> **The result:** Every project follows the same spec → plan → build → test → track pipeline. Nothing is ad-hoc. Nothing falls through cracks. Context never lost.

## Agent Behavior — Guide, Don't Enforce

The framework is **not rigid**. The AI agent adapts to how you want to work:

> **Core principle:** The user can start anywhere — jump into coding, give direct tasks, ask questions, or follow the full spec-persistent flow. All valid. The agent adapts to the user, not the other way around.

| You Do | Agent Does |
|---|---|

| | |
|---|---|
| "Build me a login page" | Adds to TASKS.md → builds it → tests it → shows results |
| "Fix this bug" | Adds to TASKS.md → fixes → marks done |
| "What should I do next?" | Walks through spec-persistent process: define specs → plan → tasks |
| "What's the status?" | Shows from TASKS.md + AGENT_CONTEXT.md |
| Come back after weeks | Reads AGENT_CONTEXT.md → summarizes where you left off |
| Never wrote specs | Retroactively fills SPECS.md from what's built — no complaints |

### Silent Tracking

Even if you never mention the process, the agent:

- Adds every task to TASKS.md before starting
- Marks tasks done when complete
- Updates AGENT_CONTEXT.md after completing work, committing, and before pushing
- Fills gaps in SPECS.md and PLANS.md retroactively
- Surfaces process naturally: "I've added this to TASKS.md", "Based on SPECS.md, we still have..."

> **User's time is sacred.** Agent does 90% of the work — writes specs, plans, tasks, tests, docs. User reviews 10%. Never ask the user to write specs or plans — the agent drafts everything, user approves or adjusts.

## File Creation/Update Rule

This single rule governs how the agent handles ALL managed files (`WORKSPACE_CONTEXT.md`, `README.md`, and all `agent/` files):

| Scenario | Action |
|---|---|
| File does not exist | Create from standard template, fill in known details |
| File exists but wrong structure | Restructure to match template — **retain all existing content**, never lose data |
| File already matches template | Leave as-is |

> **Never lose content.** When restructuring an existing file, every detail, decision, and note is preserved — just reorganized into the standard sections.

## Customization

### User Profile — Personalized AI Experience

On first session, the agent fetches your GitHub identity and asks 3 quick questions. Press Enter to use recommended, or type your own:

| Question | Options |
|---|---|
| **Communication style?** | (a) direct and concise ← RECOMMENDED<br>(b) direct, data-driven, comprehensive with tables<br>(c) conversational and collaborative<br>Press Enter to use recommended (a) |
| **Working pattern?** | (a) iterative — start small, expand ambitiously ← RECOMMENDED<br>(b) plan-first — full specs before code<br>(c) prototype-fast — build quick, refine later<br>Press Enter to use recommended (a) |
| **AI delegation?** | (a) AI does 70%, user guides 30% ← RECOMMENDED<br>(b) AI does 90%, user reviews 10%<br>(c) 50/50 collaboration<br>Press Enter to use recommended (a) |

**Saved to:**

```
~/.portable-spec-kit/user-profile/user-profile-janesmith.md         (global — asked once)
workspace/.portable-spec-kit/user-profile/user-profile-janesmith.md (committed — persists)
```

> **Personalized from first session.** Profile asked once, saved globally. On new projects: shown and confirmed — keep or customize per project. Customized profiles save to workspace. Press Enter through everything for recommended defaults.

### Your Projects

Project-specific details go in `agent/AGENT.md` (auto-created per project). This keeps the framework file portable while each project has its own stack, brand, and rules:

```
## Stack
| Layer | Technology |
|-------|-----------|
| Frontend | Next.js + TypeScript |
| Backend  | FastAPI |

## Brand
- Primary: #1B2A4A
- Accent: #4a6fa5
```

### Dev Stack Adaptation

The framework uses stack-agnostic language for testing, building, and linting. Each project's `agent/AGENT.md` defines the actual commands:

| Framework Says | Project Defines |
|---|---|
| "All tests passing" | `npm test` / `pytest` / `go test` |
| "Type checking: zero errors" | `npx tsc --noEmit` / `mypy` |
| "Linting: zero errors" | `eslint` / `ruff` / `golint` |
| "Build succeeds" | `npm run build` / `docker build` |
| "PDF generation tool" | WeasyPrint / Puppeteer / wkhtmltopdf |

> **Stack-agnostic by design.** The framework never assumes a specific language, build tool, or test runner. It defines *what* to check — each project defines *how*.

## Sharing & Syncing the Framework

### Recommended: Dedicated GitHub Repo

#### 📦 Single-Source-of-Truth Repo

One repo holds portable-spec-kit.md. All projects pull from it. Updates pushed back from any project.

```
# Setup (once)
mkdir portable-spec-kit && cd portable-spec-kit
git init && cp /path/to/portable-spec-kit.md .
git add . && git commit -m "Framework v1.0"
gh repo create portable-spec-kit --private --push

# Pull into any project
curl -sO https://raw.githubusercontent.com/you/portable-spec-kit/main/portable-spec-kit.md

# Push updates back from any project
cp portable-spec-kit.md ~/portable-spec-kit/
cd ~/portable-spec-kit && git add . && git commit -m "Updated from ProjectX" && git push
```

✓ Version controlled ✓ Pull anywhere ✓ Change history ✓ Cross-machine sync

## Other Options

| Method | How | Best For |
|---|---|---|
| **Git Submodule** | `git submodule add repo .framework` | Auto-sync across repos |
| **GitHub Gist** | `gh gist create portable-spec-kit.md` | Quick sharing, browser editable |
| **Shell Alias** | `alias init-dev='curl -sO URL/portable-spec-kit.md'` | One-command setup |

## Update Workflow Across Projects

1. **Discover new practice** while working on Project A (e.g., new testing rule)

↓

2. **AI updates portable-spec-kit.md** in Project A with the new rule

↓

3. **Push to shared repo** — single source of truth updated

↓

4. **Pull in Project B** — instantly has the new rule. AI follows it immediately.

## Portable vs Project-Specific

| In portable-spec-kit.md (Portable Framework) | In agent/AGENT.md (Per Project) |
|---|---|
| Git/push/commit rules | Tech stack choices |
| Security practices | Brand colors & fonts |
| Testing methodology | AI model/provider config |
| Naming conventions | Dev server port |
| Project templates & auto-scan | Deployment specifics |
| Code review standards | Project-specific rules |
| Spec-driven development flow | API endpoints & data model |
| Version numbering scheme | Database schema |

## Examples Included

The repo includes two example projects showing the framework in action:

| Example | What It Shows | Best For |
|---|---|---|
| `examples/starter/` | Fresh project right after setup. Self-documenting README explains every file and directory. | Understanding the framework — **start here** |
| `examples/my-app/` | Mid-development Next.js + Supabase project. 11/16 tasks done, 24 tests at 92% coverage. | Seeing a realistic project in progress |

### Starter Project Structure

```
examples/starter/
├── portable-spec-kit.md    ← Framework file (source)
├── .portable-spec-kit/     ← Kit config (user profiles)
│   └── user-profile/
├── CLAUDE.md               ← Symlink (Claude Code)
├── .cursorrules            ← Symlink (Cursor)
├── WORKSPACE_CONTEXT.md    ← Auto-created workspace state
├── README.md               ← Self-documenting (explains everything)
├── agent/
│   ├── AGENT.md           ← Stack: TBD (waiting for your specs)
│   ├── AGENT_CONTEXT.md   ← Status: "Setup — waiting for specs"
│   ├── SPECS.md           ← Empty — ready for requirements
│   ├── PLANS.md        ← Empty — ready for architecture & methodology
│   ├── TASKS.md           ← 1/5 tasks (project initialized)
│   └── RELEASES.md         ← v0.1 placeholder
```

### Mid-Development Project

```
examples/my-app/
├── agent/
│   ├── AGENT.md           ← Next.js + Supabase configured
│   ├── AGENT_CONTEXT.md   ← v0.1: 11/16 tasks, 24 tests
│   ├── SPECS.md           ← 8 features, priorities, criteria
│   ├── PLANS.md        ← Data model, APIs, 3 phases
│   ├── TASKS.md           ← 5 modules with progress
│   └── RELEASES.md         ← Categorized changelog
```

**One file. Any project. Your standards.**
The framework travels with you — consistent engineering practices across every project, every team, every AI session.

Portable Spec Kit

# Thank You

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